



SAPIENZA
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Efficient Deep Learning For Computer Vision

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List of Publications

1. F. Fontana, R. LanzinLLMo, A. Diko, G. L. Foresti, and L. Cinque. “CycleBNN: Cyclic Precision Training in Binary Neural Networks”. In: *European Conference on Computer Vision (ECCV)*. 2024.
2. F. Fontana, A. Diko, R. Lanzino, M. R. Marini, B. Kaddar, G. L. Foresti, and L. Cinque. “Revisiting Deepfake Detection: Chronological Continual Learning and the Limits of Generalization”. In: *arXiv preprint arXiv:2509.07993*. 2025.
3. F. Fontana, T. Jantos, J. Steinbrener, L. Cinque, G. L. Foresti, and B. Rinner. “BiCrossNet: Resource-Efficient Cross-View Geolocalization with Binary Neural Networks”. In: *Machine Learning: Science and Technology*. 2025.
4. R. Lanzino, F. Fontana, A. Diko, M. R. Marini, and L. Cinque. “Faster Than Lies: Real-Time Deepfake Detection using Binary Neural Networks”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024.
5. F. Fontana, A. Di Matteo, L. Cinque, G. Placidi, and M. R. Marini. “BNNAction-Net: Binary Neural Network on Hands Gesture Recognitions”. In: *ACM SIGGRAPH 2024 Posters*. 2024.
6. F. Fontana, R. Lanzino, M. R. Marini, D. Avola, L. Cinque, F. Scarcello, and G. L. Foresti. “Distilled Gradual Pruning with Pruned Fine-tuning”. In: *IEEE Transactions on Artificial Intelligence*. 2024.
7. D. Avola, L. Cinque, E. Emam, F. Fontana, G. L. Foresti, M. R. Marini, A. Mecca, and D. Pannone. “UAV Geo-Localization for Navigation: A Survey”. In: *IEEE Access*. 2024.
8. A. Diko, D. Avola, B. Prenkaj, F. Fontana, and L. Cinque. “Semantically Guided Representation Learning for Action Anticipation”. In: *European Conference on Computer Vision (ECCV)*. 2024.
9. D. Avola, L. Cinque, E. Emam, F. Fontana, G. L. Foresti, M. R. Marini, and D. Pannone. “Hand Gesture Recognition Exploiting Handcrafted Features and LSTM”. In: *International Conference on Image Analysis and Processing (ICIAP)*. 2023.
10. F. Fontana. “Training Binary Neural Networks without Floating Point Precision”. In: *arXiv preprint arXiv:2310.19815*. 2023.

Continual Learning (CL), where a model must adapt to new deepfake generators as they emerge over time. By designing a novel chronological evaluation framework that mirrors the real-world historical progression of generative models, the research uncovers a fundamental limitation in generalization. While CL methods prove effective at efficiently adapting to new generators without forgetting old ones, they consistently fail to generalize to future, unseen generators. This empirical finding gives rise to the *Non-Universal Deepfake Distribution Hypothesis*, which posits that each generator imprints a unique, non-transferable signature on its creations. This conclusion has significant implications, suggesting that continuous adaptation is not just an efficiency measure but an absolute necessity for any deepfake detection system intended for long-term, real-world deployment.

The thesis culminates with its most ambitious application, a task that synthesizes the demands for extreme efficiency and high performance in a complex, safety-critical domain. The study, BiCrossNet: Resource-Efficient Cross-View Geolocalization with BNNs [49], applies the BNN methodology to the problem of matching images from a UAV to geo-referenced satellite imagery, a crucial capability for navigation in GNSS-denied environments. The dramatic shifts in viewpoint, scale, and illumination make this a formidable matching problem. To succeed, the research develops a suite of specialized training techniques, including a Bi-Gradual Unfreezing schedule to stabilize transfer learning and a Bi-Partitioned Optimization strategy that applies different optimizers to the full-precision and binary components of the network. The resulting model delivers a remarkable efficiency-performance trade-off, achieving results competitive with much larger models while reducing the required computational operations by over 90 times. This final demonstration, enabling on-board, real-time geolocalization for a UAV, serves as the capstone evidence for the thesis’s central argument. By progressing from core methods to a series of increasingly demanding applications, this body of work collectively demonstrates that computational efficiency is not a secondary concern or a mere engineering optimization, but rather a key that unlocks the potential for advanced artificial intelligence to operate in the dynamic, constrained, and ever-evolving physical world.